

What AI systems can and cannot do

Module 1 - Lesson 1

Overview

AI systems recognize patterns and make predictions from data. This lesson sets realistic expectations: what modern AI excels at, where it fails, and why understanding limits matters.

Key Concepts

- Most modern AI is statistical pattern matching trained on large datasets.
- AI excels at perception, language, and prediction where patterns exist at scale.
- Models are narrow: a model trained for one task cannot generalize to unrelated tasks.
- Hallucinations, bias, and fragility are failure modes of pattern matching, not bugs to ignore.
- Correlation learned from data is not causation; models reflect their training data.
- Human oversight is required for high-stakes decisions.

Example

Use cases where AI is strong:

- Detecting spam, translating language, recommending content
- Image and speech recognition
- Predicting churn, fraud, and demand

Where AI is weak:

- Common-sense reasoning, planning, long-horizon goals
- Rare edge cases absent from training data
- Tasks requiring genuine understanding of cause and effect

Summary

AI is powerful pattern matching, not intelligence. It thrives on large, patterned datasets and struggles with reasoning, novelty, and causation. Know the difference before applying it.

Practice Checklist

- List three tasks where AI would help and three where it would mislead.
- Identify a failure mode (bias, hallucination) in a real AI product.
- Write one sentence describing what your model actually learns from data.